

DNS Walkthrough

What actually happens between typing a web address and a page loading

A website's address, like **example.com**, is something a human can read, but computers don't talk to each other using names — they use numbers, called IP addresses (something like 104.21.32.10). DNS (the Domain Name System) is the system that translates the name someone types into the actual number the computer needs to connect to. It's often described as the internet's phone book: you look up a name, and it gives you the number to call.

What happens, step by step

1. You type an address. Say you type **cevdetsatar.netlify.app** into your browser and hit enter. Your computer doesn't know the number for that name yet, so it asks for help.

2. The resolver takes the question. Your computer sends the request to a **resolver** — usually run by your internet provider or a public service like Google's or Cloudflare's. Think of the resolver as a helpful assistant whose whole job is to go find the answer to “what's the number for this name?” on your behalf.

3. The resolver asks a nameserver. The resolver doesn't just know every website's number by memory. It asks a **nameserver** — a server that's responsible for keeping the official records for a domain. Nameservers are organized in a hierarchy (starting from very general ones down to the specific one that knows about, say, netlify.app), and the resolver works its way down that chain until it reaches the nameserver that actually holds the answer for the exact address being asked about.

4. The nameserver returns a record. The nameserver holds a small database of **records** — entries that map a name to something useful. The most common type, an **A record**, maps a name directly to an IP address. A **CNAME record** is slightly different: instead of pointing straight to a number, it points to another name, and says “whatever this other name resolves to, use that.” This is exactly what happens when you connect your own domain to a host like Netlify: Netlify gives you a name like **your-site.netlify.app**, and your own domain's CNAME record simply says “for www.yourdomain.com, go look up your-site.netlify.app instead, and use whatever address that gives you.” That way, if Netlify ever changes which physical server hosts your site, you never have to update your own domain's records — the CNAME just keeps pointing at Netlify's name, and Netlify handles the rest.

5. The resolver answers your computer. Once the resolver has the final IP address (having followed any CNAME chain to the end), it sends that number back to your browser.

6. Your browser connects directly to that address. With the actual IP address in hand, your browser opens a connection straight to the server at that address and asks for the page. The server sends the page back, and that's what you see load on screen.

Why this matters for a site like mine

When I deployed my site on Netlify, Netlify automatically gave me a working name (something like **random-name-123.netlify.app**) that already has real DNS records pointing to Netlify's servers — that's why the site was reachable immediately, with no DNS setup required on my part. Netlify also handles HTTPS automatically: the padlock in the browser comes from a certificate Netlify issues and manages behind the scenes, so I didn't have to request or install one myself. If I later buy my own domain (like cevdetsatar.com), the only thing I'd need to do is add a CNAME record at wherever I bought the domain, pointing it at my Netlify site's name — DNS would then take care of routing anyone who types my custom domain to the exact same place my current Netlify URL already goes.

For now, no custom domain is connected — the site is live on its free Netlify subdomain, which is enough for this assignment.